CS341 · COMPUTER GRAPHICS

Midterm 472 Practice

20 marks

Muharib

01 Questions

Q1 SHORT ANSWER · 5 PTS

1. Why use aspect ratio?
2. What is the difference between absolute and relative coordinates?
3. Define a vector.
4. Write the formula to compute a vector from point P to point Q.
5. Rays emitted by a cathode ray tube ..... light .....

Q2 TRUE / FALSE · 4 PTS

1. A vector has both magnitude and direction. ( )
2. Sum two points results in a valid point. ( )
3. The dot product of two perpendicular vectors is equal to zero. ( )
4. A unit vector has a length greater than 1. ( )

Q3 MCQ · 4 PTS

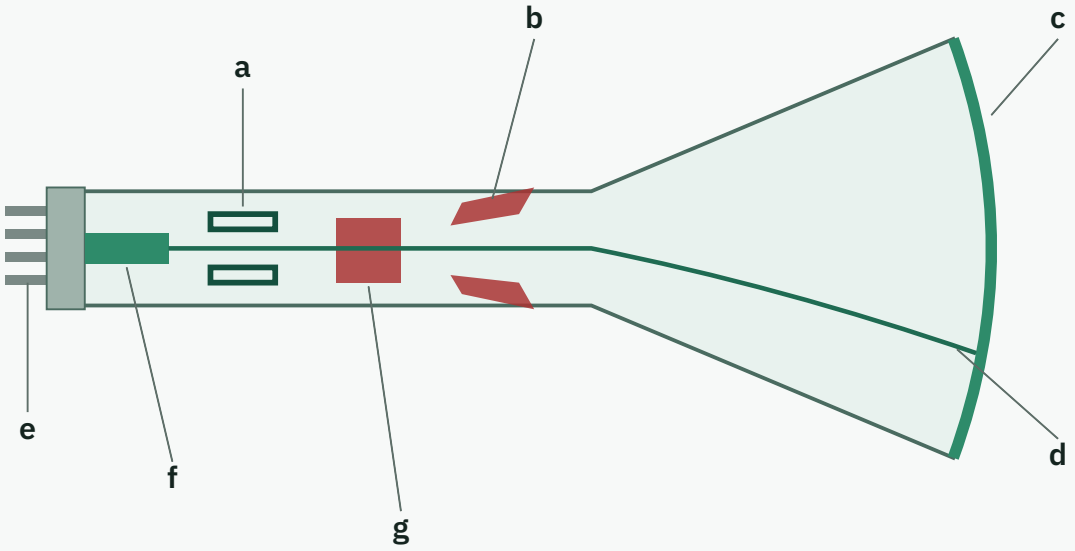
1. In raster scan display, picture information is stored in:
   A) Vector list B) Frame buffer C) CPU registers D) Refresh file
2. Which of the following display methods draws lines directly using an electron beam?
   A) Raster scan display B) Random scan display C) Plasma display D) LCD display
3. What does the clarity of a display screen depend on?
   A) Memory size B) CPU speed C) Dot pitch D) Hard disk capacity
4. Which of the following is an input device?
   A) Monitor B) Printer C) Mouse D) Speaker

Q3 MCQ · CONT.

5. Which of the following is NOT a main task for computer graphics?
   A) Modeling B) Rendering C) Animation D) Data encryption
6. One of the applications of computer graphics:
   A) Computer Aided Design (CAD) B) Medical Visualization C) Entertainment D) All of the above
7. What is the main function of computer graphics?
   A) Creating, storing, and manipulating models and images
   B) Managing hardware components
   C) Processing numerical calculations only
   D) Controlling network communication
8. What is computer graphics?
   A) Creating, storing, and manipulating models and images
   B) Managing hardware components
   C) Processing numerical calculations only
   D) Controlling network communication

Q4 CRT LABELS · 3 PTS

Label the cathode ray tube using the following letter components:



Q5 DDA · 4 PTS

Based on the DDA algorithm, calculate the points between the starting point (2,2) and ending point (4,10). Then, draw the line using these points.

k	x	y	round(x)	round(y)
0				
1				
2				
3				
4				
5				
6				
7				
8				

